Problem 2. Find a least-squares sol-n of the system $Ax = b$, where

(1) $$A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}.$$

Use any method.

Sol-n:

(i) First easy way to do this Problem might be by normal equations, A^TA; given

$$Ax = b \quad \text{f. } A^T \text{ by left}$$

(2) $$A^T \cdot A\hat{x} = A^T b$$

Claim is the solution $\hat{x}$ for normal equation in (2) is the least-squares solution for (1) as $R(A^TA) \equiv R(A)$.

Thus

(3) $$A^TA = \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix}^T \cdot \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -2 & 2 \end{bmatrix} \quad \&$$

$$A^Tb = \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix}^T \cdot \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

Then

$$A^TA\hat{x} = A^Tb \quad \text{becomes}$$

(4) $$\begin{bmatrix} 3 & -2 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \Rightarrow$$

(5) $$\hat{x} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$